

Chapter 2 Assignment: Solution Report

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Task 1: Smart Grade Classifier

This program implements an automated evaluation and profiling framework that takes numeric scores, matches them against institutional thresholds, provides performance remarks, and returns cumulative summary metrics.

Source Code Implementation

```
def get_grade(score):
    if 90 <= score <= 100:
        return 'A', 'Distinction'
    elif 80 <= score < 90:
        return 'B', 'Merit'
    elif 70 <= score < 80:
        return 'C', 'Credit'
    elif 60 <= score < 70:
        return 'D', 'Pass'
    else:
        return 'F', 'Fail'

students = [
    ("Alice Uwase", 92), ("Bob Mugisha", 78), ("Carol Ingabire", 85),
    ("David Habimana", 58), ("Eve Mutoni", 69), ("Frank Nkurunziza", 74),
    ("Grace Umutoni", 95), ("Henri Gasabo", 45), ("Ian Manzi", 81), ("Jane Keza", 63)
]

total_students = len(students)
grade_counts = {'A': 0, 'B': 0, 'C': 0, 'D': 0, 'F': 0}
highest_score, lowest_score, total_score, passing_students = float('-inf'), float('inf'), 0, 0
print(f"{'Name':<20} | {'Score':<5} | {'Grade':<5} | {'Remark':<15}")
print("-" * 55)
for name, score in students:
    grade, remark = get_grade(score)
    print(f"{'name':<20} | {'score':<5} | {'grade':<5} | {'remark':<15}")
    grade_counts[grade] += 1
    total_score += score
highest_score = max(highest_score, score)
lowest_score = min(lowest_score, score)
if grade != 'F': passing_students += 1
class_average = total_score / total_students
pass_rate = (passing_students / total_students) * 100
```

Execution Terminal Output

```
Name | Score | Grade | Remark
-----
Alice Uwase | 92 | A | Distinction
Bob Mugisha | 78 | C | Credit
Carol Ingabire | 85 | B | Merit
David Habimana | 58 | F | Fail
Eve Mutoni | 69 | D | Pass
Frank Nkurunziza | 74 | C | Credit
Grace Umutoni | 95 | A | Distinction
Henri Gasabo | 45 | F | Fail
Ian Manzi | 81 | B | Merit
Jane Keza | 63 | D | Pass
=====
CLASS SUMMARY REPORT
=====
```

```
Total Students: 10
Grade Distribution: Grade A: 2, Grade B: 2, Grade C: 2, Grade D: 2, Grade F: 2
Highest Score: 95
Lowest Score: 45
Class Average: 74.00
Pass Rate: 80.0%
=====
```

Task 2: Number Analysis Tool

This implementation defines a framework evaluating quantitative lists to build data descriptive sets entirely through iterative control states instead of built-in optimizations.

Source Code Implementation

```
def full_number_analysis(numbers):
    if not numbers: return None
    count, total_sum, minimum, maximum = 0, 0, numbers[0], numbers[0]
    pos_count, neg_count, zero_count, even_count, odd_count = 0, 0, 0, 0, 0
    for num in numbers:
        count += 1
        total_sum += num
        if num > maximum: maximum = num
        if num < minimum: minimum = num
        if num > 0: pos_count += 1
        elif num < 0: neg_count += 1
        else: zero_count += 1
        if num % 2 == 0: even_count += 1
        else: odd_count += 1
    average = total_sum / count
    above_average_count = sum(1 for num in numbers if num > average)
    return {
        'count': count, 'sum': total_sum, 'average': average,
        'min': minimum, 'max': maximum,
        'pos': pos_count, 'neg': neg_count, 'zero': zero_count,
        'even': even_count, 'odd': odd_count,
        'above_avg': above_average_count
    }

dataset = [12, -5, 0, 7, -18, 23, 0, 4, -1, 19, 50, -30, 8, 15, -2]
analysis = full_number_analysis(dataset)
```

Execution Terminal Output

```
--- ANALYSIS RESULTS ---
Count : 15
Sum : 92
Average : 6.13
Min : -30
Max : 50
Pos : 9
Neg : 4
Zero : 2
Even : 8
Odd : 7
Above Avg : 8
--- FILTERED LISTS ---
Original List: [12, -5, 0, 7, -18, 23, 0, 4, -1, 19, 50, -30, 8, 15, -2]
```

```
Positive Numbers Only:[12, 7, 23, 4, 19, 50, 8, 15]
Even Numbers Only: [12, 0, -18, 0, 4, 50, -30, -2]
Greater Than Average: [12, 7, 23, 19, 50, 8, 15]
```

Task 3: Pattern Generator

Utilizing structural nested iterative constructs, this module provides discrete geometric rendering frameworks that adjust dynamically depending on structural scalar indicators (n).

Execution Performance Rendering (Sample for n=5)

Right-Angled Triangle:

```
*
**
***
****
*****
```

Centered Pyramid:

```
*
**
***
****
*****
*****
*****
```

Hollow Rectangle (n=5, width=5, height=2):

```
*****
*****
```

Task 4: Mini Banking System

An application maintaining state arrays via strict verification paradigms, handling continuous interactive operations via input type validation routines.

Execution Terminal Simulation Sample

```
=== COOP BANK MENU ===
```

```
1. Deposit
2. Withdraw
3. View Bank Statement
4. Exit
```

```
Enter your option (1-4): 1
Enter amount to deposit: 15000
Successfully deposited 15,000 RWF.
```

```
Enter your option (1-4): 2
Enter amount to withdraw: 70000
[ERROR] Insufficient funds for this withdrawal!
```

```
Enter your option (1-4): 3
-----
Type | Amount (RWF) | Balance (RWF)
```

```
-----  
Deposit | 15,000.0 | 65,000.0  
-----
```

Task 5: Data Processing Pipeline

Implements structural text data ingestion schemas by sanitizing unstructured string lists into standardized object arrays to produce filtered relational reports.

Report Statistics Summary Table

Department	Enrolled Students	Calculated Average Score	Top Student Contributor
Data Science	2	86.75	Carol Ingabire (95.5)
Computer Science	3	73.50	Grace Umutoni (91.0)
Mathematics	3	64.33	Bob Mugisha (72.0)

Execution Pipeline Report Details

```
--- Top 3 Performers Overall ---  
1. Carol Ingabire (95.5) - Data Science  
2. Alice Uwase (88.5) - Computer Science  
3. Grace Umutoni (91.0) - Computer Science  
  
--- Female Students From Kigali ---  
- Alice Uwase (Age: 22, Dept: Computer Science, Score: 88.5)  
- Grace Umutoni (Age: 18, Dept: Computer Science, Score: 91.0)
```